AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 13 · STUDENT LESSON

Batching and epochs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Week 12 printed states numbered 0 to 4. How many parameter updates occurred?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Distinguish an example a batch an update and an epoch. Try the recall before opening notes.

Your learning target

I can distinguish an example a **batch** an **update** and an **epoch**.

I can trace parameter changes across successive **batch**es.

I can calculate **update** counts and aggregate unequal **batch** losses correctly.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Distinguish an example a **batch** an **update** and an **epoch**

Trace parameter changes across successive **batch**es

Calculate **update** counts and aggregate unequal **batch** losses correctly

Retrieve what you already know

> Week 12 printed states numbered 0 to 4. How many parameter **update**s occurred?

My starting explanation

This week's end goal

By the end, I can explain **batch**ing and **epoch**s using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Batching and epochs case. Separate observations, given facts and assumptions before choosing an explanation.

In Week 12 every **update** used both **training** examples. We now keep the same network h = ReLU(x), y = vh, with only v trained, and divide the data into **batch**es. In this lesson each nonempty **batch** produces exactly one **update**.

An **epoch** visits every **training** example exactly once, without replacement. These rules make the counts unambiguous.

Imagine five fictional records A through E in fixed order. With **batch size** 2, the groups are [A, B], [C, D] and [E]. Keep the smaller last **batch**.

One **epoch** has three **batch**es and three **update**s; two **epoch**s have six **update**s and ten example visits.

Training rows	Batch size	Batch lengths	Updates per epoch
5	2	2, 2, 1	3
5	5	5	1
5	1	1, 1, 1, 1, 1	5

When the last **batch** is retained, **update**s per **epoch** equal the row count divided by **batch size** rounded up. A program that drops the final incomplete **batch** follows a different rule and omits examples in that pass.

Record the rule instead of silently changing the denominator.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Batching and epochs. Connect each labelled input, deciding step and result.

Use the two fictional pairs A = (1, 2) and B = (2, 4), start v = 0 and rate 0.1. For one pair with hidden value h, squared-error gradient is 2 times (vh - target) times h.

For a **batch**, average these gradients at the same current v. Apply one **update**, then recompute for the next **batch**.

```
def epoch(v, rows, size, rate):
    if not rows or size < 1:
        raise ValueError("Need rows and positive batch size")
    trace = []
    for start in range(0, len(rows), size):
        batch = rows[start:start + size]
        hs = [max(0.0, x) for x, target in batch]
        gradient = sum(2 * (v * h - target) * h
                       for h, (x, target) in zip(hs, batch)) / len(batch)
        v = v - rate * gradient
        trace.append(v)
    return trace

rows = [(1.0, 2.0), (2.0, 4.0)]
print([round(v, 6) for v in epoch(0.0, rows, 1, 0.1)])  # [0.4, 1.68]
print([round(v, 6) for v in epoch(0.0, rows, 2, 0.1)])  # [1.0]
```

Assume finite numeric pairs, integer size and finite rate. With size 1, A gives gradient -4 and new v = 0.4. B then uses 0.4, giving **prediction** 0.8, residual -3.2 and gradient -12.8, so v becomes 1.68.

With size 2, both gradients are evaluated at v = 0; their mean is -10, producing v = 1.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Batching and epochs. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Use only one full-**batch update** after reversing record order.

Predicted result

Prediction: The **update**d weight remains 0.2.

Why the result follows

Reason: The same mean of the two starting gradients is unchanged by their order.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Batching and epochs to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Batch	A group of training examples used together to calculate one gradient here
Update	One application of the gradient and learning rate to the current parameters
Epoch	One pass through all training examples under the stated sampling rule
Batch size	The intended number of examples in each batch with a possible smaller final batch

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Batching and epochs, predict a step, then compare with its explanation.

One pass can contain different numbers of updates

BATCH RULE	PARAMETER AFTER EACH BATCH	UPDATES
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

For B then A separately, show both gradients and new parameter values. Why does one epoch still contain two updates?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Batching and epochs

Show why an **epoch**'s example order can matter for single-example **update**s.

The facts we will use

Supplied facts

Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/13

Batching and epochs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. A then B

A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08.

2. Restart and reverse

B at w=0 has gradient 0; A then gives w = 0.4.

3. Compare full batch

At w = 0 mean gradient=(-4+0)/2=-2; one full-batch update gives w = 0.2.

CHANGE ONE THING

Use only one full-batch update after reversing record order.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. A then B	A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08.
2. Restart and reverse	B at w=0 has gradient 0; A then gives w = 0.4.
3. Compare full batch	At w = 0 mean gradient=(-4+0)/2=-2; one full-batch update gives w = 0.2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

What this example does not establish

Keep the stated limit

Different order effects do not mean a universally better **batch**ing strategy; compare

declared schedules and budgets fairly.

Keep the stated limit

Read the labelled picture: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Use only one full-**batch update** after reversing record order. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Batching and epochs claim and explain a repair using the supplied evidence.

Check the claim against the facts

Batch losses observed during **training** can refer to different parameter values and different examples. They are not automatically the full **training** loss of the final model.

To compare final models, freeze each one and evaluate every **training** example again. Here full **training** MSE is 2.5 times (v - 2) squared: at v = 1 it is 2.5, and at v = 1.68 it is 0.256. These are **training** results only.

Check the claim against the facts

Even at one fixed model, averaging **batch** means without their sizes gives the wrong dataset mean when **batch**es have different lengths. Suppose the squared errors on A to E are [1, 1, 4, 4, 9].

Batch means for lengths 2, 2, 1 are [1, 4, 9]. Full MSE is (2 times 1 + 2 times 4 + 1 times 9) / 5 = 3.8, not (1 + 4 + 9) / 3. Use example counts as weights.

Locate the earliest unsupported step

At one frozen model, **batch**es of sizes 3 and 1 have mean squared errors 2 and 10. Find the full MSE and explain why the average of 2 and 10 is wrong.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Batching and epochs with support, then explain the deciding step in your own words.

For 10 rows, **batch size** 4 and three **epoch**s, retaining the final **batch**, list **batch** lengths and count total **update**s and example visits.

For B then A separately, show both gradients and new parameter values. Why does one epoch still contain two updates?

Explain your prediction to a partner before comparing results.

Practise with less help: Repair a stale gradient

Use workbook W2 for the connected example "**Batch**ing and **epoch**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Batch**ing and **epoch**s. Proposed explanation: "One **epoch** always means one **update**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Batching and epochs. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

```
def epoch(v, rows, size, rate):
    if not rows or size < 1:
        raise ValueError("Need rows and positive batch size")
    trace = []
    for start in range(0, len(rows), size):
        batch = rows[start:start + size]
        hs = [max(0.0, x) for x, target in batch]
        gradient = sum(2 * (v * h - target) * h
                       for h, (x, target) in zip(hs, batch)) / len(batch)
        v = v - rate * gradient
        trace.append(v)
    return trace

rows = [(1.0, 2.0), (2.0, 4.0)]
print([round(v, 6) for v in epoch(0.0, rows, 1, 0.1)])  # [0.4, 1.68]
print([round(v, 6) for v in epoch(0.0, rows, 2, 0.1)])  # [1.0]
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Batching and epochs. Explain why each step is justified.

Explain → apply → challenge

For 10 rows, **batch size** 4 and three **epoch**s, retaining the final **batch**, list **batch** lengths and count total **update**s and example visits.

At one frozen model, **batch**es of sizes 3 and 1 have mean squared errors 2 and 10. Find the full MSE and explain why the average of 2 and 10 is wrong.

A program **process**es 11 rows using size 4 and drops the last incomplete **batch**. How many examples did it visit and how many **update**s occurred? What must its report disclose?

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **batch**ing and **epoch**s **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish an example a **batch** an **update** and an **epoch**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Equal **epoch**s mean equal example visits under our rule, but changing **batch size** changes the number of **update**s. A comparison should report both counts, starting values, order, learning rate and the rule for the last **batch**.

Neither smaller **batch**es nor more **update**s guarantee better independent performance.

> **Two teams train on five rows for two epochs. One uses size 2 and the other size 5, retaining every row. Are their example visits and update counts equal? Explain what to record.**

Next week holds the **batch** rule fixed while comparing learning rates.

> A program **process**es 11 rows using size 4 and drops the last incomplete **batch**. How many examples did it visit and how many **update**s occurred? What must its report disclose?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Batching and epochs case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

One pass can contain different numbers of updates

BATCH RULE	PARAMETER AFTER EACH BATCH	UPDATES
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Test order on a new case independently

Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values. Compare the final values.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values. Compare the final values.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Batching and epochs, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Show why an **epoch**'s example order can matter for single-example **update**s.

By the end, I can	How I will show it
Distinguish an example a batch an update and an epoch	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Trace parameter changes across successive batches	Explain the deciding step, show a trace or test, and state one limit.
Calculate update counts and aggregate unequal batch losses correctly	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Batching and epochs without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish an example a **batch** an **update** and an **epoch**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Show why an **epoch**'s example order can matter for single-example **update**s.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.

1. Restart and reverse

2. A then B

3. Compare full **batch**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare full **batch** At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Use only one full-**batch update** after reversing record order. Which conclusion is supported by the supplied facts?

1. The **update**d weight remains 0.2.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **update**d weight remains 0.2. The same mean of the two starting gradients is unchanged by their order.

Repair a missing building block

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and **update** steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

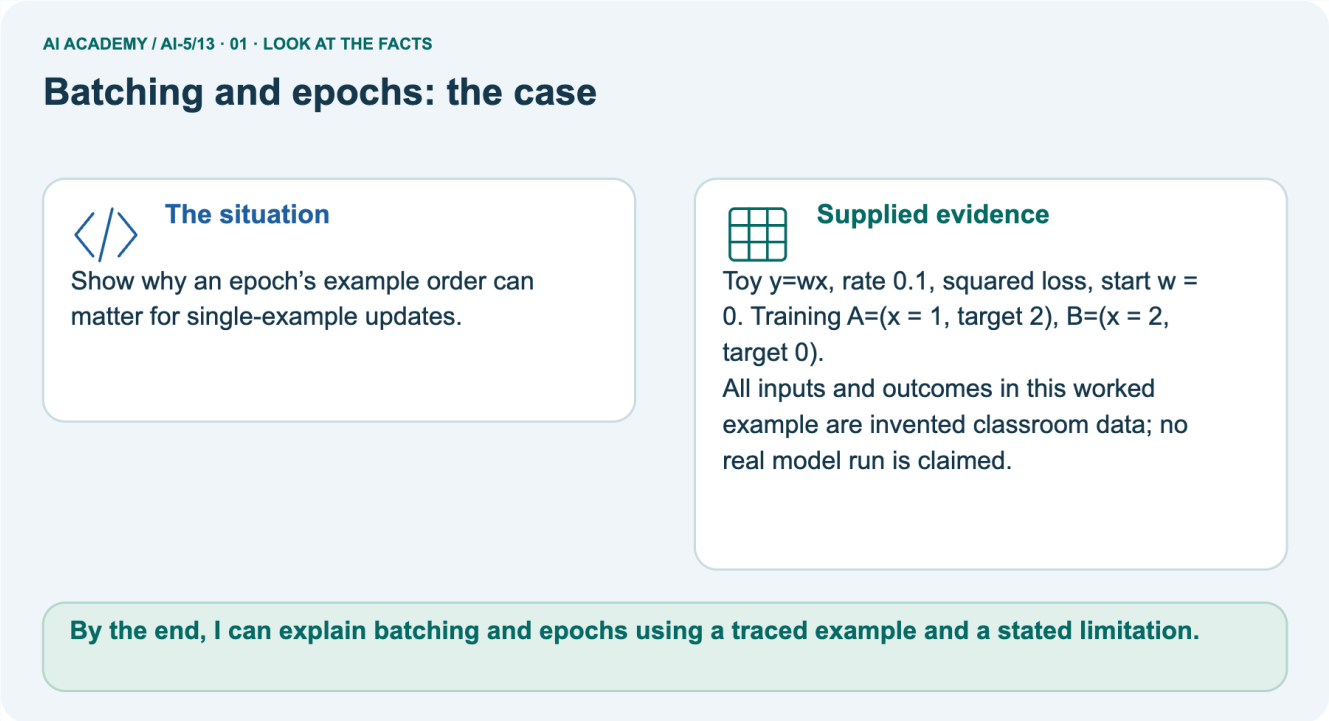
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Batching and epochs graphic using its labels, deciding evidence and relevant limitation.

01. Batching and epochs: the case



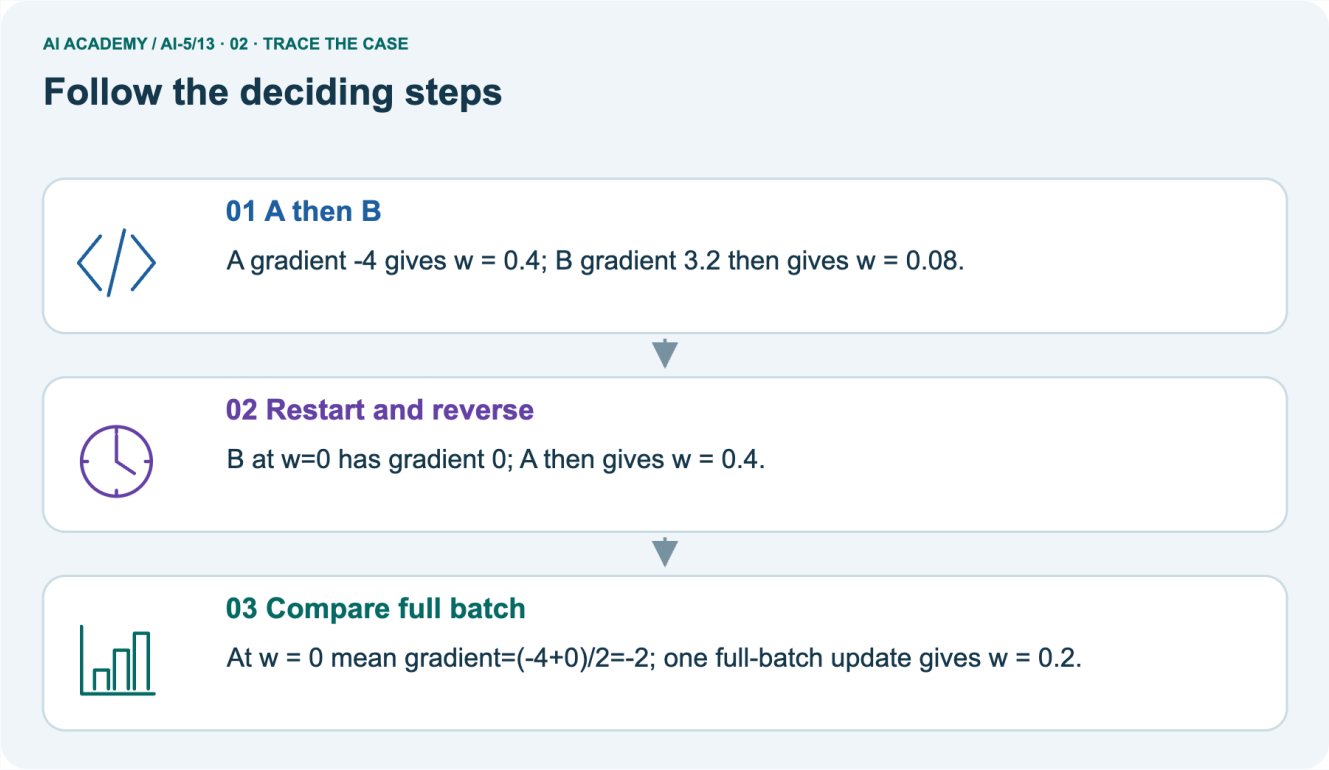
Follow the labelled picture with the matching explanation. Open the vector diagram

Show why an **epoch**'s example order can matter for single-example **update**s. Supplied facts: Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

A then B: A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08. → Restart and reverse: B at w=0 has gradient 0; A then gives w = 0.4. → Compare full **batch**: At w = 0 mean gradient= (-4+0)/2=-2; one full-**batch update** gives w = 0.2.

> Point to each numbered stage and explain which recorded **input** it uses.

03. One pass can contain different numbers of updates

AI ACADEMY / AI-5/13 · 03 · INSPECT THE LESSON MODEL

One pass can contain different numbers of updates

Batch rule	Parameter after each batch	Updates
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

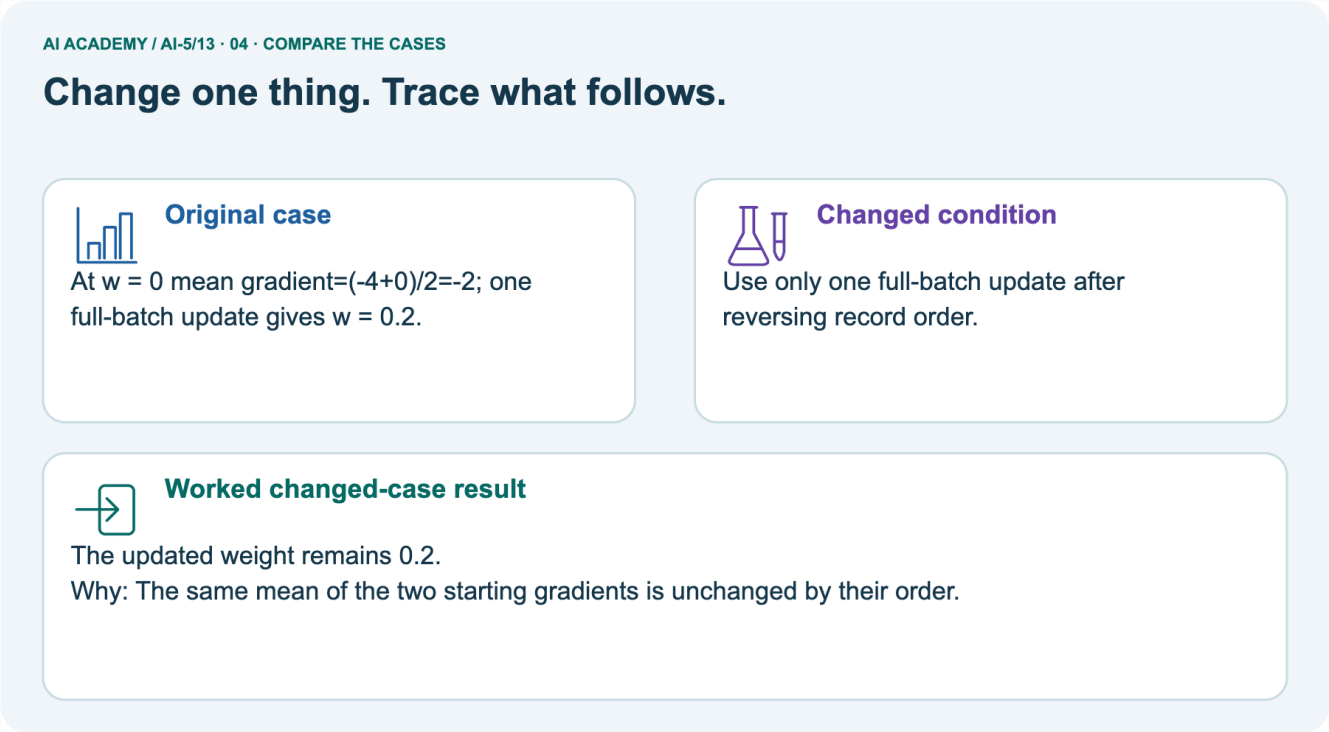
All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

One pass can contain different numbers of **update**s. A then B separately | 0 to 0.4 to 1.68 | 2. A and B together | 0 to 1 | 1. B then A separately | 0 to 1.6 to 1.68 | 2

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



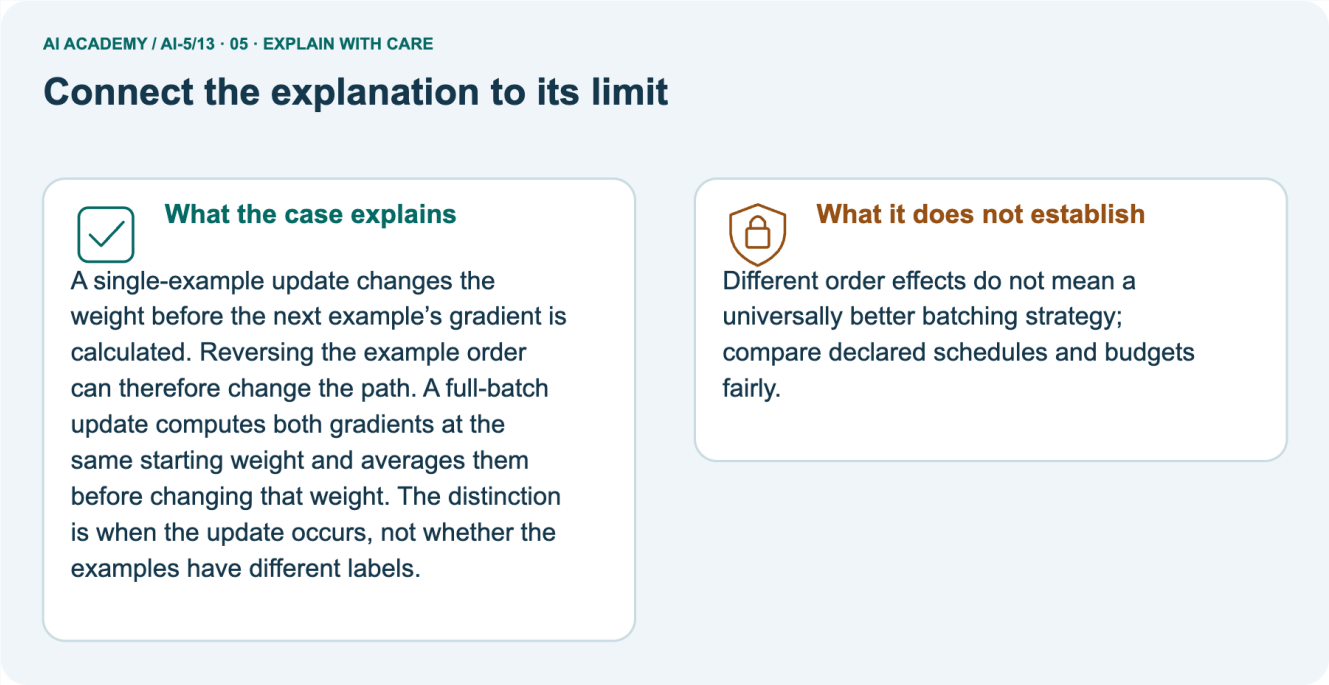
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2. Changed condition: Use only one full-**batch update** after reversing record order. Expected result: The **update**d weight remains 0.2.

Reason: The same mean of the two starting gradients is unchanged by their order.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

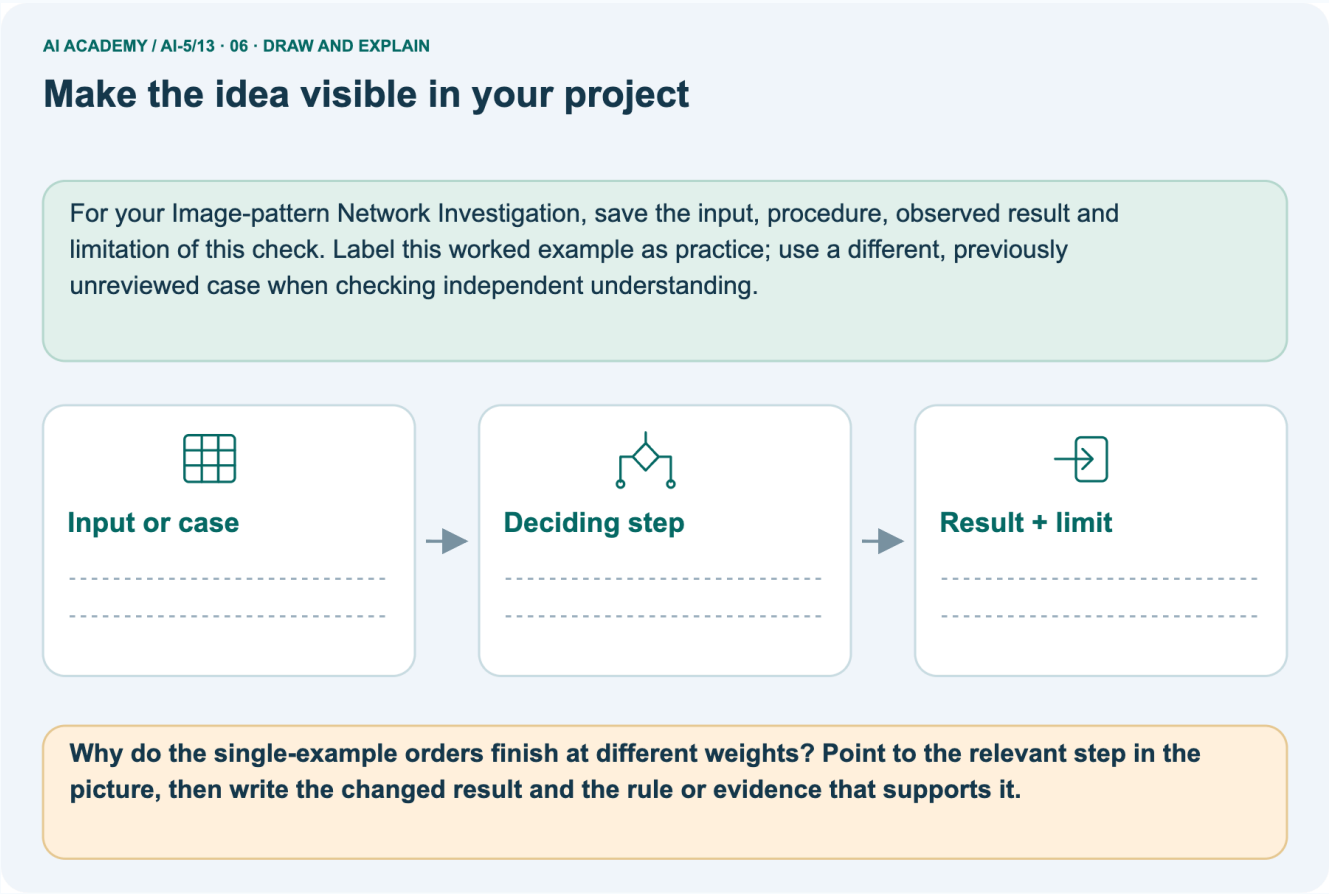
A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Limit: Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.